

How the Proof Works

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Finite kinetic energy does not rule out concentration. An incompressible vortex can stretch along its axis only by narrowing across it. If r is the core radius and U is the change in velocity across that core, then

$$|\nabla u| \sim \frac{U}{r}, \quad \int_{B_r} |u|^2 dx \sim U^2 r^3.$$

Shrinking r raises the velocity gradient through the factor $1/r$, while the energy carried by that region can remain finite because its volume falls like r^3 . The energy bound leaves that possibility open, so the equation has to tell us what happens as the scale shrinks:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0.$$

Transport can sharpen the profile, while viscosity diffuses it. Rescaling a concentrating profile shows how both effects change as the profile narrows. For $\lambda > 1$, set

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

The spatial width is divided by λ , and the time scale is divided by λ^2 . With $\Phi_\lambda(x, t) = (\lambda x, \lambda^2 t)$, the chain rule gives

$$\begin{aligned} \partial_t u_\lambda &= \lambda^3 (\partial_t u) \circ \Phi_\lambda, & (u_\lambda \cdot \nabla) u_\lambda &= \lambda^3 ((u \cdot \nabla) u) \circ \Phi_\lambda, \\ \nabla p_\lambda &= \lambda^3 (\nabla p) \circ \Phi_\lambda, & \nu \Delta u_\lambda &= \lambda^3 (\nu \Delta u) \circ \Phi_\lambda. \end{aligned}$$

All four parts grow by λ^3 . A narrower profile strengthens nonlinear transport and viscous diffusion together; scaling alone does not give either one control of the small scales.

Under this rescaling,

$$\begin{aligned} \widehat{u}_\lambda(\xi, t) &= \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t), \\ \|u_\lambda(t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} |\widehat{u}(\xi/\lambda, \lambda^2 t)|^2 d\xi \\ &= \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2. \end{aligned}$$

For $s < \frac{1}{2}$, the norm falls as the profile narrows. For $s > \frac{1}{2}$, it rises. At $s = \frac{1}{2}$, it is unchanged. So $s = \frac{1}{2}$ is critical: its norm neither grows nor shrinks as the profile narrows.

At that height, the critical measurement stays unchanged when the rescaling is repeated. Set $\lambda = 2$. Halving the core radius then quarters the corresponding time scale:

$$r_n = 2^{-n}r_0, \quad \delta t_n = 4^{-n}\delta t_0, \quad \sum_{n=0}^{\infty} \delta t_n = \frac{4}{3}\delta t_0.$$

The series shows how infinitely many scale reductions could fit inside a finite time. A singularity would require one solution to pass through every scale in that sequence.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

If a singularity were forming, each higher time derivative should show less spatial regularity, not more. The differentiated equations reverse that expectation.

$$\partial_t u = \nu \Delta u - (u \cdot \nabla)u - \nabla p,$$

$$\partial_t^2 u = \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla)\partial_t u - \nabla \partial_t p,$$

$$\partial_t^3 u = \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u - 2(\partial_t u \cdot \nabla)\partial_t u - (u \cdot \nabla)\partial_t^2 u - \nabla \partial_t^2 p.$$

For $n \geq 0$,

$$\begin{aligned} \partial_t^{n+1} u &= \nu \Delta \partial_t^n u \\ &\quad - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u \\ &\quad - \nabla \partial_t^n p. \end{aligned}$$

Set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p.$$

At the critical height, that temporal sequence appears inside one energy. Write

$$H(t) := \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2, \quad \Lambda := (-\Delta)^{1/2}.$$

For any $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2.$$

Let $\mathcal{P}_s$ denote the transport-pressure contribution at height s . Pairing the viscous term at that height gives

$$\nu \operatorname{Re} \langle \Lambda^s \Delta u, \Lambda^s u \rangle_{L^2} = -\nu \|\Lambda^{s+1} u\|_2^2 = -2\nu E_{s+1}.$$

Adding $\mathcal{P}_s$ gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

This is where viscosity moves the calculation from height s to height $s+1$, with coefficient 2ν .

With $H = E_{1/2}$,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2},$$

$$H'' = \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2},$$

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

At order $n \geq 1$,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The n -th derivative of H reaches $E_{n+1/2}$. As temporal order rises, Sobolev height rises with it:

$$\text{higher temporal order} \quad \longrightarrow \quad \text{higher spatial Sobolev order}.$$

Control of every finite height through T_* would give, by Sobolev embedding,

$$s > k + \frac{3}{2} \quad \implies \quad H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3),$$

and hence

$$\bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Fix finite temporal order N and finite Sobolev height M . At temporal order r ,

$$E_{r,s}(t) := \frac{1}{2} \|\Lambda^s V_r(t)\|_2^2,$$

$$\partial_t E_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}.$$

Stopping at $r = N$ and $s = M$ leaves finitely many adjacent rows. Their total velocity norm $\mathfrak{R}_{N,M}(u; T)$ stays bounded as $T \uparrow T_*$:

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty \quad (N, M < \infty).$$

Adding the corresponding $Q_0, \dots, Q_N$ coordinates gives the pressure-complete rectangle $\mathcal{R}_{N,M}[u_0; T]$.

Whenever two rectangles overlap, their common entries are derivatives of u and p already fixed by the differentiated equations, so the overlaps agree. Their compatible intersection is

$$\mathcal{I}[u_0] := \left(\mathcal{R}_{N,M}[u_0] \right)_{N,M < \infty},$$

and every finite pair of indices occurs in one of those blocks. Hence

$$u \in \bigcap_{r,m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

Projecting onto $r = 0$ gives

$$\mathcal{S}_0 := \pi_0 \mathcal{I}[u_0].$$

For every finite spatial height,

$$u \in C([0, T_*]; H^m(\mathbb{R}^3)).$$

Thus

$$u_* := u(T_*) \in \bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Local existence gives a smooth solution starting from u_* , and uniqueness makes it agree with u on their overlap. The solution passes through T_* , contradicting finite maximality. Thus

$$T_* = \infty.$$

For a finite L^p norm of $\nabla^k \partial_t^r u$, choose an integer m with

$$m > k + \frac{3}{2} - \frac{3}{p}, \quad 2 \leq p \leq \infty.$$

Since $\partial_t^r u \in C_t H_x^m$, Sobolev embedding gives

$$\nabla^k \partial_t^r u \in L_t^\infty L_x^p.$$

Choosing $m = 1, 2, 3$ gives

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^1 \longrightarrow L_t^\infty L_x^6,$$

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^2 \longrightarrow L_t^\infty L_x^\infty,$$

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^3 \longrightarrow L_t^\infty W_x^{1,\infty}.$$

For $m = 3$,

$$\begin{aligned} \int_0^{T_*} \|\nabla u(t)\|_\infty dt &\leq T_* \sup_{0 \leq t \leq T_*} \|\nabla u(t)\|_\infty \\ &\leq CT_* \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^3} < \infty. \end{aligned}$$

In shorthand,

$$\begin{aligned} u_0 &\longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \\ &\longrightarrow \mathcal{I}[u_0] \longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty. \end{aligned}$$

The estimates come from the intersection, not the other way around.